



Amir Hossein Madadi

B.Sc. Electrical Engineering — Power Systems

Final-year Electrical Engineering student at the University of Tehran, specializing in power systems, electrical machines, transformers, and engineering simulation. Proficient in computational analysis, algorithm implementation in MATLAB, and digital system design with Verilog HDL.

Contact

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Technical Skills

Power Systems & Machines

- MATLAB
- Simulink
- Ansys Maxwell
- Simcenter SPEED
- COMSOL
- DlgSILENT PowerFactory
- PowerWorld Simulator

Circuits, Electronics & Design

- PSpice
- LTspice
- Multisim
- OrCAD Capture
- Altium Designer
- AutoCAD
- PCB Design

Digital, Control & Programming

- Verilog HDL
- ModelSim
- Quartus Prime
- Python
- C / C++
- Siemens PLC

Languages

Persian: Native
English: Advanced
Arabic: Intermediate

Interests

Electrical machines & transformers
Power grid analysis & operation
Motor control & dynamic simulation
Digital design & processor architecture

Seeking internship or collaboration opportunities in electrical engineering projects.

Education

B.Sc. in Electrical Engineering

2021 — Present

University of Tehran, Faculty of ECE — Power Systems

Selected Projects

Power System Analysis & Load Flow

- Load flow and short-circuit studies on the IEEE 39-Bus test network using DlgSILENT PowerFactory, including maximum loadability, generator outage scenarios, and voltage profiling.
- N-1 Contingency analysis and three-phase fault current studies on the test network.
- 400kV/230kV transmission line parameter analysis on real tower configurations and network operation under various loading conditions.
- Implementation of Newton-Raphson, Fast Decoupled, and Gauss-Seidel algorithms in MATLAB for convergence comparison on large networks.

DlgSILENT PowerFactory

MATLAB

IEEE 39-Bus

Load Flow

Short Circuit

High-Voltage Insulation & Electric Field Simulation

- 2D electric potential and field distribution analysis between electrodes using Finite Difference Method (FDM) and Monte Carlo simulation in MATLAB.
- Effect of multiple dielectric layers with varying permittivity and air voids analyzed using COMSOL.
- Traveling wave simulation on 300Ω overhead lines (1500km/400km) with damping behavior studied in LTspice.

COMSOL

MATLAB

LTspice

FDM

Monte Carlo

Electrical Machines & Transformer Design

- Design and construction of a high-frequency transformer using EE55 ferrite core, including core selection, litz wire for skin-effect mitigation, and practical validation.
- Three-phase 7.5kW/380V induction motor simulation in Ansys Maxwell-RMxprt with stator/rotor characterization and torque-speed analysis.
- DC motor design and optimization in Simcenter SPEED including geometric and electromagnetic parameter calculation.

Ansys Maxwell

RMxprt

Simcenter SPEED

COMSOL

Control Systems & Dynamic Modeling

- DC shunt motor speed control implementation and performance analysis in Simulink/Simscape.
- Physical modeling of a single-loop DC motor including magnetic field, electromagnetic force, and induced EMF in MATLAB.
- CSTR (irreversible exothermic A→B reaction) modeling and control with 3 coupled differential equations in Simulink.

Simulink

Simscape

MATLAB

CSTR

FPGA Digital System Design

- Direct Digital Synthesis (DDS) function generator on FPGA with DAC-PWM, generating sine, square, triangular, sawtooth, half-wave, and full-wave rectified waveforms.
- Radix-4 multiplier and Exponential Engine SoC accelerator implementation in SystemVerilog.
- Fundamental FPGA circuits including ring oscillator, frequency divider, and Baud rate generator.

Verilog

Quartus Prime

FPGA

SystemVerilog

RISC-V Processor Design

- Design and simulation of three RV32I-class processor architectures (single-cycle, multi-cycle, and 5-stage pipeline) in Verilog HDL.
- Control unit, 32-register file, 32-bit ALU, and instruction/data memory implementation.
- Data and control hazard management: Forwarding, Stall, Flush for 5-stage pipeline.
- Validation with RISC-V assembly programs including 32-bit signed array sorting.

Verilog HDL

ModelSim

Quartus Prime

RISC-V